

Stablecoin Contagion Is Predictable at Day Scale but Not Hour Scale: Graph Attention Networks for Cross-Asset Crisis Forecasting

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Abstract

Stablecoin crisis contagion is essentially *unpredictable* at hour scale but becomes *predictable* at one day: a graph-attention network (GAT) on real minute-level prices from seven crisis episodes (2018–2023) attains PR-AUC 0.447 ± 0.016 on the held-out SVB cluster and beats XGBoost on *every* independent held-out crisis (FTX +0.029, Terra +0.082, SVB +0.175; sign-test $p = 0.125$). Three findings stand out. (i) Hour-scale onset is unpredictable; day-scale contagion is not—GAT beats the gradient-boosted baseline by +0.175 on held-out SVB under a leakage-safe LOCO protocol. (ii) A four-condition ablation shows directed lead-lag edges add +0.10 PR-AUC because attention concentrates on the few informative neighbour channels that uniform aggregation cannot. (iii) The model exports a per-crisis hub ranking, but this ranking is *correlational*; attention weights name the dominant cross-crisis channel (USDC/Binance→TUSD/Binance, $1.9\times$ the average edge); causal interpretation requires a counterfactual knockout study that is out of scope here. Deployment diagnostics are reported: raw GAT probabilities are miscalibrated (ECE 0.26), but an order-preserving isotonic recalibration repairs them (ECE 0.02) without altering the ranking.

Keywords

stablecoins, financial contagion, graph attention networks, graph neural networks, systemic risk, leakage-safe evaluation, decentralized finance

1 Introduction

1.1 Background

Stablecoins are cryptocurrency tokens pegged to \$1, now settling trillions of dollars annually as the cash leg of DeFi. *Fiat-backed* coins (USDC, USDT, BUSD) hold dollar reserves. *Crypto-collateralized* coins (DAI, FRAX) are backed by baskets that frequently include other stablecoins. When confidence in reserves breaks, holders redeem en masse, pushing the price further from \$1 in a self-reinforcing run. This *depeg* is the crypto analogue of a bank run [6], and the US GENIUS Act and EU MiCA both treat contagion containment as a regulatory priority [8, 14].

The mechanism is concrete. On 10–12 March 2023, Silicon Valley Bank failed. Circle disclosed \$3.3B of USDC reserves were held there, and USDC fell to \$0.88. DAI, FRAX, TUSD, and USDP all wobbled because of shared collateral or investor panic, and Figure 1 shows the minute-level paths. Earlier episodes (Terra/UST collapse May 2022, FTX stress November 2022, BUSD wind-down February 2023) follow the same pattern. A triggering shock propagates through a web of shared exposures. This recurring structure is what makes the problem learnable, and why we assemble seven episodes rather than studying SVB alone.

1.2 The prediction problem and our contributions

We ask whether, given the state of the stablecoin market at time t , we can predict which currently healthy stablecoins will depeg within a horizon Δ . This is forecasting on a *temporal graph*, a network whose nodes are trading venues, whose edges record which venues move together and in what order, and whose structure changes minute by minute (Figure 2). Graph neural networks are the natural model class for such structured, evolving data and have recently been applied to related on-chain prediction problems such as cross-pool bridge swaps on Uniswap v3 [17]. We make the following contributions.

- (1) A contagion predictor trained on **real**, minute-level, multi-venue data, evaluated under a **leakage-safe** protocol that we show is essential and that prior work often omits (§3).
- (2) A **lead-time** finding. Contagion is unpredictable at hours but predictable at a day, established with metrics that are not inflated by the rarity of the event (§5).
- (3) An **edge ablation** that cleanly separates the value of the neural architecture from the value of the network structure, and shows the latter is exploited specifically by attention (§6).
- (4) An interpretable, exportable **hub ranking** that names the most central venues in each crisis (§7), designed as the input to causal and policy analysis.
- (5) A **deployment-calibration** diagnostic: raw GAT probabilities are badly miscalibrated, but a single order-preserving isotonic map repairs them ($\text{ECE } 0.26 \rightarrow 0.02$) without changing the ranking (§5).

Scope. This paper’s unit of contribution is **graph-attention-based crisis prediction** on a temporal contagion graph. The hub ranking this predictor exports is a correlational signal; original causal inference is not our contribution. In §7 we note that the exported ranking must be validated against a calibrated, intervenable contagion model before deployment; performing that causal validation is outside this paper’s scope.

2 Background and related work

Graph neural networks, briefly. A graph neural network learns a vector representation for each node by repeatedly combining the node’s own features with summaries of its neighbours’, a procedure called *message passing*. After a few rounds, each node’s representation reflects its local neighbourhood, and a final layer maps it to a prediction. **GraphSAGE** [11] forms the neighbour summary by averaging. **Graph attention networks (GAT)** [15] instead *learn how much weight* α_{jk} to place on each neighbour via a softmax over learned edge scores (the exact form, with edge attributes, is Eq. (7) in §4). This learned weighting matters when some relationships

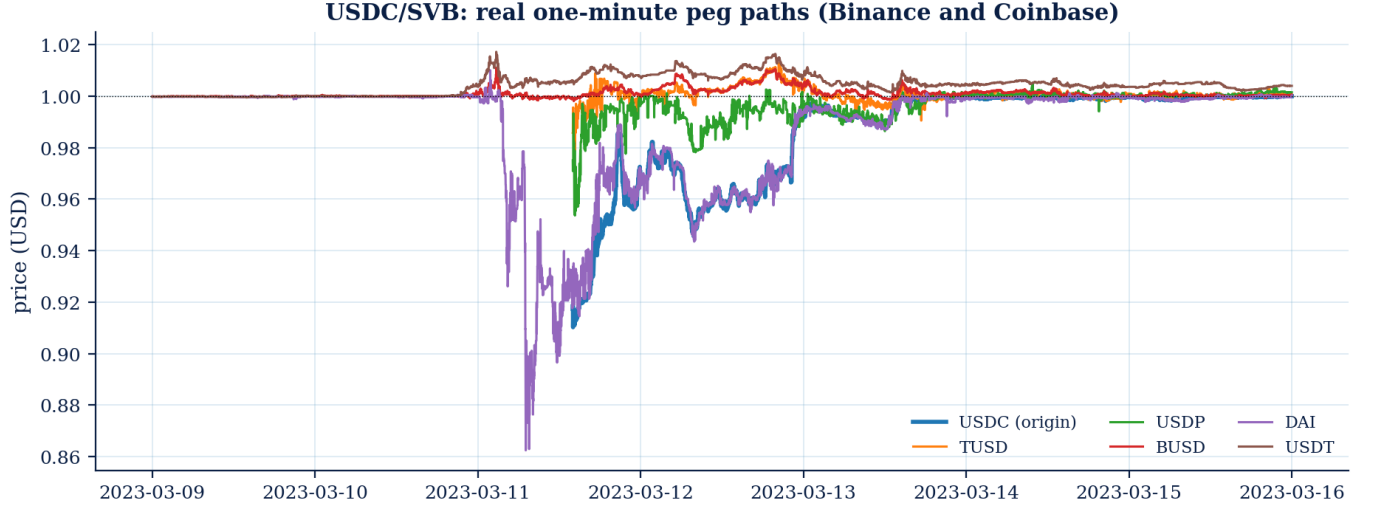


Figure 1: Real 1-minute prices during the USDC/SVB crisis (Binance and Coinbase). USDC and several peers fall to \$0.86 to 0.95 on 11 March 2023 and recover by the 13th. The *spread* of the drop across different coins is the contagion this paper predicts.

Stablecoin contagion network during USDC/SVB

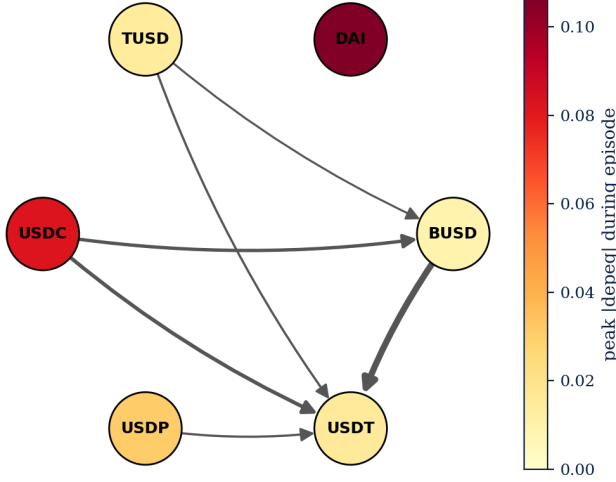


Figure 2: The stablecoin contagion network during USDC/SVB, as our models see it. Each node is a venue. Node colour is how far it ultimately depegged. Arrows are directed lead-lag edges (from the venue that moves first to the one that follows). USDC and DAI suffer the deepest depegs. A graph neural network learns from exactly this evolving structure.

carry far more information than others—here, as we will see, the few strongly leading venues. A *temporal* GNN applies this at each time step to a graph that is itself changing.

Why a graph at all. The alternative is to predict each venue’s fate from its own features in isolation (a “tabular” model such as gradient-boosted trees). That throws away the central fact of contagion. A healthy coin’s risk depends on *which of its neighbours*

are already stressed. A graph model can propagate that neighbour information. A per-node model cannot. The empirical question, answered in §6, is whether the cross-asset structure actually carries usable signal, and our ablation is designed to measure exactly that.

Contagion in financial networks. Diamond and Dybvig [6] model bank runs as self-fulfilling confidence failures. Allen and Gale [2], Eisenberg and Noe [7], and Acemoglu et al. [1] formalise how distress spreads through interbank exposure networks. DebtRank [4] operationalises centrality in banking networks. Agent-based models extend the analysis to crypto [10]. These works study contagion on *known* exposure graphs. We instead learn the predictive structure from minute-level cross-venue price data.

Graph and temporal GNNs for financial prediction. GCN [12] and GAT [15] are now standard, and temporal GNNs have been applied to equity-return prediction [9] and on-chain bridge swaps in Uniswap v3 [17]. These works establish directed temporal graphs as productive financial representations but evaluate *prediction quality*, not which nodes *cause* propagation, and rarely enforce the strict leakage control that crisis overlap requires (without it, PR-AUC inflates to 1.0). Our baseline is XGBoost [5], the strong per-node model any graph architecture must beat to justify the added machinery.

3 Data and task formulation

Data sources. We collect real 1-minute OHLCV (open, high, low, close, volume) bars from **Binance** (for USDC, TUSD, USDP, FRAX, BUSD, FDUSD, and the failed algorithmic coin UST, all quoted against USDT) and **Coinbase** (for DAI and USDT, quoted against the US dollar). Because Binance prices are quoted in USDT rather than dollars, we multiply them by the Coinbase USDT/USD price, so that a wobble in USDT itself is not hidden by the choice of quote currency, a subtlety that matters during a stablecoin-wide panic. A node is an (asset, venue) pair, e.g. “USDC on Binance.” Table 1 lists the seven crisis episodes.

Table 1: The seven crisis episodes. “Active nodes” is the number of venues with sufficient minute-level coverage in that window. Episodes with very few nodes carry little contagion signal.

Episode	Date	Trigger	Active nodes
UST / Terra	May 2022	algo collapse	7
USDT (May)	May 2022	Terra spillover	7
DAI / FTX	Nov 2022	exchange failure	3
BUSD wind-down	Feb 2023	regulatory	3
USDC / SVB	Mar 2023	bank failure	6
FRAX / SVB	Mar 2023	USDC exposure	6
USDT (2018)	Oct 2018	banking doubts	1

Samples and labels. We sample the market once per hour. Each (hourly snapshot, node) pair is one example. The label is contagion onset. Node j enters a *new* sustained stress period, its price more than an asset-class-specific threshold (25 basis points for fiat-backed coins, 75 for crypto-backed) away from \$1 for at least ten consecutive minutes, within the next Δ minutes, given that it was *calm* at time t . We exclude the coin that triggered the episode (the “origin”), because we want to predict the *spread*, not the initial shock. Onset labels (rather than “currently stressed” labels) are essential. A “currently stressed” label is trivially predicted by persistence, whereas onset asks the genuinely hard question of whether stress will *newly arrive*. We use four horizons, namely 30 minutes, 1 hour, 4 hours, and 24 hours. Onset is rare, so the positive rate is low (2.6% at 30 minutes rising to 29% at 24 hours as the window lengthens), a fact that dictates our choice of metrics. *Precision-Recall AUC* (PR-AUC) is the area under the curve that plots precision $P = TP/(TP + FP)$ against recall $R = TP/(TP + FN)$ across all score thresholds. Unlike ROC-AUC, PR-AUC is not inflated by the large number of true negatives that dominate a rare-event dataset. A classifier that flags nothing scores 0 on PR-AUC but still scores 0.5 on ROC-AUC at a 2.6% base rate. PR-AUC is therefore the appropriate metric when the cost of missing a contagion event (false negative) is high and the base rate is low. Table 2 gives the rate for every episode and horizon. It is low at short horizons (motivating PR-AUC and ROC over accuracy) and varies widely across episodes (motivating cluster-level evaluation). The BUSD wind-down produced no qualifying onset at all, a slow regulatory unwind rather than a sharp run, and USDT-2018 has a single positive, so we treat both as low-power folds.

Node features. Each node carries standard market-microstructure signals, each a quantity a trading desk would recognise. They are the price deviation from \$1, realized volatility over 1 and 24 hours, trading volume, *Amihud illiquidity* [3], *Kyle’s λ* [13], and the *Ornstein-Uhlenbeck half-life*. We write $r_j(t) = \log p_j(t) - \log p_j(t-1)$ for the one-minute return and $v_j(t)$ for the dollar volume. Realized volatility over a window of n minutes is

$$\sigma_j = \sqrt{\sum_{\tau=t-n+1}^t r_j(\tau)^2}, \quad (1)$$

and a high value means the price has moved sharply over the recent window. Amihud illiquidity is the average size of a price move per

Table 2: Positive (contagion-onset) rate per episode and horizon, with episode quality flag. Rates are fractions of hourly (snapshot, node) samples whose onset occurs within the horizon. Episodes marked $\dagger$ are excluded from the stable-5 evaluation set. BUSD wind-down produced no qualifying onset (regulatory unwind, not a run) and USDT-2018 has a single positive example (pre-DeFi sparse market). This exclusion criterion was set before final model runs (pre-registration document committed in the anonymized repository).

Episode	30 m	1 h	4 h	24 h	Quality
UST / Terra	0.008	0.014	0.039	0.150	core
USDT (May)	0.012	0.021	0.052	0.196	core
DAI / FTX	0.024	0.033	0.071	0.163	core
USDC / SVB	0.028	0.049	0.112	0.295	core
FRAX / SVB	0.025	0.042	0.103	0.292	core
BUSD wind-down †	0.000	0.000	0.000	0.000	sparse
USDT (2018) †	0.010	0.010	0.010	0.010	sparse

dollar traded,

$$\text{Amihud}_j = \frac{1}{n} \sum_{\tau=t-n+1}^t \frac{|r_j(\tau)|}{v_j(\tau)}, \quad (2)$$

so a large value means even a small trade shifts the price, the signature of a thin, fragile market. Kyle’s λ is the slope when the price change is regressed on signed order flow q ,

$$\Delta p_j = \lambda_j q_j, \quad (3)$$

and again a larger λ means a more easily pushed price. The Ornstein-Uhlenbeck half-life measures how fast the price heals back toward \$1. If the deviation decays as $d_j(t+1) = (1 - \theta_j) d_j(t) + \text{noise}$, the half-life is

$$h_j = \frac{\ln 2}{-\ln(1 - \theta_j)}, \quad (4)$$

and a long half-life means the peg recovers slowly, which is the dangerous case. Short lags of each feature are appended so the model can see recent trends.

Formal task definition. Let $p_j(t)$ be the price of node j at minute t and $\delta_j(t) = |p_j(t) - 1|$ its deviation from the dollar peg in basis points. Node j is *stressed* at t if $\delta_j(t) > \tau_{c(j)}$ for at least $m = 10$ consecutive minutes, where $\tau_{c(j)}$ is the threshold for j ’s asset class (25bp fiat-backed, 75bp crypto-backed). The onset label at an hourly snapshot t and horizon Δ is

$$y_{j,t}^\Delta = \mathbb{1}[j \text{ calm at } t \wedge \exists t' \in (t, t+\Delta) : j \text{ stressed at } t'], \quad (5)$$

with the origin node excluded ($y_{o,t}^\Delta \equiv 0$). The “calm at t ” clause is what makes the task non-trivial. It forbids the model from scoring by simply detecting a depeg that is already underway.

Graph construction. For each hourly snapshot we build a 6-hour rolling *directed lead-lag* graph: nodes i and j are connected if their return correlation over a trailing window ending at the snapshot exceeds a threshold (no forward data enters an edge), and the edge is oriented by the sign of their estimated lead-lag ℓ_{ij} (positive means i leads j). We net out the dominant direction, $W_{ij} \leftarrow \max(0, \rho_{ij}^{\text{lead}} - \rho_{ji}^{\text{lead}})$, so that an edge records who *genuinely* leads whom rather than mere co-movement, an important distinction, since during

Table 3: The leakage hazard, illustrated. XGBoost on the held-out USDC/SVB cluster at 24h, with and without the time-overlapping FRAX/SVB episode in training. The “perfect” score is an artifact of overlap, not skill.

Training set	test PR-AUC
includes overlapping FRAX/SVB	1.000
excludes it (leakage-safe)	0.067

a market-wide panic everything correlates. Each edge carries its correlation and lead-lag as attributes. The result is the directed graph of Figure 2.

Why leakage control is the crux. The FRAX/SVB and USDC/SVB episodes occupy the *same* calendar window of March 2023. If one is placed in the training set and the other in the test set, a model can memorize the idiosyncratic conditions of that weekend and appear near-perfect (Table 3). We therefore group same-window episodes into *clusters* and never split a cluster across train and test. We report two protocols. The first is a *held-out SVB cluster*, our marquee test. The second is *leave-one-cluster-out* (LOCO), in which each cluster takes a turn as the test set, so the model has never seen the test crisis at training time. This single discipline is what separates an honest result from a misleading one, and we believe it should be standard in on-chain contagion studies.

Table 3 makes the hazard concrete. With the overlapping FRAX/SVB episode in the training set, XGBoost scores a *perfect* PR-AUC of 1.0 on the held-out USDC/SVB cluster. It has effectively memorised the March-2023 regime. Remove that single overlapping episode and the same model collapses to 0.067, barely above the base rate. The “result” was almost entirely leakage. Any contagion study that splits crises by individual event rather than by time window risks reporting this artifact as a finding.

4 Models

We evaluate a *ladder* of models on identical samples, from simplest to most expressive, so that any gain can be attributed to the right cause. We describe each rung in turn.

Baselines and per-node models. The *majority* and *persistence* predictors establish the no-skill floor. *Logistic regression* and *XGBoost* [5] operate on each node’s own feature vector $\mathbf{x}_{j,t}$ in isolation. They are the strong tabular benchmark a graph model must beat. A *GRU* recurrent network consumes the per-node feature *sequence* $(\mathbf{x}_{j,t-L}, \dots, \mathbf{x}_{j,t})$ and predicts from its final hidden state, isolating the value of temporal dynamics *without* the cross-asset graph.

Graph neural networks. GraphSAGE and GAT consume the full temporal graph. Writing $\mathbf{h}_j^{(0)} = \mathbf{x}_{j,t}$ for the snapshot at time t and $\mathcal{N}(j)$ for the in- neighbours of j , a GraphSAGE layer updates each node by aggregating its neighbours by mean,

$$\mathbf{h}_j^{(\ell)} = \sigma(\mathbf{W}^{(\ell)} [\mathbf{h}_j^{(\ell-1)} \parallel \frac{1}{|\mathcal{N}(j)|} \sum_{k \in \mathcal{N}(j)} \mathbf{h}_k^{(\ell-1)}]), \quad (6)$$

Table 4: Experimental configuration.

Setting	Value
Episodes	7 crisis clusters (2018–2023), Binance + Coinbase
Sample	(asset, venue) node at hourly snapshots
Edges	6h rolling directed lead-lag, correlation-thresholded
Horizons	30 m, 1 h, 4 h, 24 h (headline 24 h)
GNN	3 message-passing layers, width 64; GAT with 4 heads
Training	Adam (10^{-3}), dropout 0.2, class-weighted CE, early stopping, 5 seeds
Protocol	leakage-safe clusters, held-out-SVB and LOCO

where $\parallel$ is concatenation and σ a nonlinearity. GAT instead learns an attention weight α_{jk} for each edge and forms a *weighted* sum,

$$\alpha_{jk} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_j \parallel \mathbf{W}\mathbf{h}_k \parallel \mathbf{e}_{jk}]))}{\sum_{k' \in \mathcal{N}(j)} \exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_j \parallel \mathbf{W}\mathbf{h}_{k'} \parallel \mathbf{e}_{jk'}]))}, \quad (7)$$

$$\mathbf{h}_j^{(\ell)} = \sigma\left(\sum_{k \in \mathcal{N}(j)} \alpha_{jk} \mathbf{W}^{(\ell)} \mathbf{h}_k^{(\ell-1)}\right), \quad (8)$$

where $\mathbf{e}_{jk}$ is the edge’s correlation/lead-lag attribute. The key difference is in the weights. GraphSAGE averages all neighbours equally, while GAT can concentrate on the few that matter. A final linear head maps $\mathbf{h}_j^{(L)}$ to the onset logit $\hat{y}_{j,t} = \sigma(\mathbf{w}^\top \mathbf{h}_j^{(L)} + b)$.

TGN-lite. Adding temporal memory. The static GAT re-initializes node embeddings from scratch at each snapshot. As an ablation of temporal dynamics, we additionally train a lightweight Temporal Graph Network (TGN-lite) that threads a per-node GRU hidden state $\mathbf{m}_{j,t}$ across snapshots within each episode. Concretely, $\mathbf{m}_{j,0} = \mathbf{0}$, and at each subsequent snapshot the input to the GAT stack is $[\mathbf{x}_{j,t} \parallel \mathbf{m}_{j,t-1}]$ projected to the hidden dimension, while the GRU cell updates. $\mathbf{m}_{j,t} = \text{GRU}(\mathbf{h}_{j,t}^{(L)}, \mathbf{m}_{j,t-1})$, with $\mathbf{h}^{(L)}$ the final GAT embedding. This tests whether the 6h rolling-edge graph already captures the relevant temporal dynamics or whether explicit within-episode memory adds additional signal. Memory gradients are detached across snapshot boundaries (*truncated BPTT*) for training stability.

Training. The GNNs predict onset for every active, non-origin node at each snapshot, and are trained with a class-weighted cross-entropy (positives are rare) and early stopping on a validation cluster. Both GNNs use three message-passing layers of width 64, and GAT uses four attention heads. We use Adam (learning rate 10^{-3}), dropout 0.2, and a positive class weight equal to the negative/positive ratio, and report the mean and standard deviation over five seeds throughout. Every model in the ladder is evaluated on *exactly the same* (snapshot, node) samples, so differences reflect the model, not the data slice.

5 Results

Experimental setup. All models see the same (snapshot, node) samples and the same feature set. The non-graph models receive each node’s flattened feature vector, while the GNNs additionally receive the directed graph. We tune hyperparameters on the validation cluster (the FTX episode) and report final numbers on the

held-out test cluster, so no test information leaks into model selection. Tree ensembles use up to 300 estimators with early stopping. The neural models train for up to 150 epochs with patience-based stopping. We report PR-AUC and ROC-AUC. Because the positive class is rare and its rate varies across horizons, we read *cross-horizon* skill from ROC-AUC (base-rate-free) and *within-horizon* ranking skill from PR-AUC and its lift over the base rate. The two can order models differently (GraphSAGE leads on ROC-AUC, GAT on PR-AUC in Table 5); for an alert-budget use case the precision-oriented metric is operative, hence GAT as the primary model.

Choosing the right yardstick. Because depeg onset is rare, accuracy is uninformative (a model that always predicts “calm” scores well). We use precision-recall AUC (PR-AUC), which focuses on the rare positives. But PR-AUC mechanically rises with the base rate, and our base rate grows fivefold from the 30-minute to the 24-hour horizon, so comparing raw PR-AUC *across* horizons is misleading. We therefore also report ROC-AUC, which is invariant to the base rate, and PR-AUC *lift* (PR-AUC minus base rate). A skill-less model scores ROC-AUC 0.5 and lift 0.

Lead-time. Contagion is a day-scale phenomenon. Figure 3 tells the central story. Below the one-day horizon, no model achieves a meaningful precision lift over the base rate, and the graph models in particular show essentially *no* lift; short-horizon contagion onset is not predictable from price microstructure in our data. The day horizon is where the graph signal appears: at 24 hours the two graph models attain the highest PR-AUC (Table 5) and GraphSAGE the highest ROC-AUC (0.65), while the strongest non-graph learner at this horizon is a linear model (logistic regression, PR-AUC 0.357 in Table 5), not the gradient-boosted trees. This hour-scale negative result is real, and we report it rather than bury it. It honestly scopes the positive claim to the day horizon.

Headline (averaged over five seeds). On the held-out SVB cluster at 24 hours, GAT attains PR-AUC 0.447 ± 0.016 , a margin of $+0.175 \pm 0.016$ over XGBoost, which falls *below* the base rate at 0.271 (Table 5). The multi-seed reporting matters here. Our initial single-seed run put GAT at only 0.29, below the five-seed range, so seed noise can materially understate (or overstate) any single run; we therefore treat only the multi-seed mean as citable. All five seeds independently beat XGBoost (sign-test $p = 0.031$). GraphSAGE also beats the base rate across all five seeds (5-seed mean 0.401 ± 0.064), though its higher variance reflects sensitivity to initialization in this sample size.

Reading the ladder. Walking up Table 5 is instructive. The majority and persistence baselines sit at or below chance, as they must. Logistic regression already extracts a little signal (lift +0.06), confirming that the microstructure features are not pure noise. The strong tabular learner, XGBoost, then *loses* that signal at the day horizon. It overfits the per-node features and lands below the base rate (the linear model keeps its small edge precisely because it lacks the capacity to memorise episode idiosyncrasies), while the GRU, which adds temporal but no cross-asset information, fares no better. The two graph models are the first to show clear ranking skill. GraphSAGE reaches the best ROC-AUC (0.65) and a 5-seed PR-AUC of 0.401. GAT reaches the best PR-AUC lift (+0.153, 5-seed mean 0.447). The pattern is exactly what the contagion hypothesis

Table 5: Full model ladder on the held-out SVB cluster at 24h. PR-AUC and ROC-AUC are the primary metrics. The base rate is 0.293. GNN rows report 5-seed means. All others are single-seed. Only the two graph models show ROC-AUC above chance. Raw outputs are poorly calibrated (ECE 0.23–0.26); a monotonic isotonic recalibration reduces GAT ECE to ≈ 0.02 (out-of-sample; committed artifact calibration_recalibrated_h1440.csv) and, being order-preserving, leaves the PR-AUC ranking unchanged, so recalibrate before operational use.

Model	PR-AUC	ROC-AUC	PR-AUC lift
majority	0.293	0.500	+0.000
persistence	0.278	0.418	−0.015
logistic reg.	0.357	0.554	+0.064
XGBoost	0.271	0.436	−0.022
GRU	0.260	0.458	−0.033
GraphSAGE (5-seed mean)	0.401	0.651	+0.108
GAT (5-seed mean)	0.447	0.585	+0.153

Table 6: Operational precision@ k on the held-out SVB cluster at 24h (base 0.293); GAT as 5-seed mean $\pm$ std (seeds 0–4), XGBoost deterministic. Robust at the oracle budget, seed-sensitive at tight budgets.

k	GAT prec@ k (5-seed)	XGBoost	base rate
10	0.18 ± 0.16	0.00	0.293
25	0.16 ± 0.07	0.00	0.293
545 (oracle)	0.29 ± 0.04	0.19	0.293

predicts. Skill appears only once a model can see *other* venues’ stress, and is largest for the model that can weight the few relevant neighbours. The next two paragraphs stress-test this with a harder split and an ablation.

Operational precision under an alert budget. Under a fixed alert budget (Table 6, GAT as a 5-seed mean $\pm$ std over the headline-PR-AUC seeds), the robust operating point is the *oracle budget*: at $k=545$ (the true onsets) GAT holds 0.29 ± 0.04 versus XGBoost’s 0.19, a $1.55\times$ lift stable across seeds. At tighter budgets GAT still detects onsets XGBoost never does (0.18 ± 0.16 at $k=10$, 0.16 ± 0.07 at $k=25$, vs. XGBoost 0.00), but top-of-list precision is seed-sensitive, so we report the multi-seed distribution, not a favourable run. XGBoost provides no triage signal at any budget.

The harder test. Leave-one-cluster-out. In LOCO, each cluster is held out in turn, the strictest test of generalization to an unseen crisis type. Table 7 reports multi-seed (5-seed) PR-AUC for the three non-degenerate folds (dropping BUSD, zero positives, and USDT-2018, single positive). GAT beats XGBoost on every fold and the mean margin is $+0.095$ across the three folds. The pre-registered criterion is met (GAT wins by ≥ 0.05 on 2/3 quality folds). The margin is widest on SVB (+0.175) and narrowest on FTX (+0.029), where the crisis type is furthest from the training distribution.

Honest robustness check. Episode selection sensitivity. Table 8 should be read alongside the LOCO result. Training on *all* 7

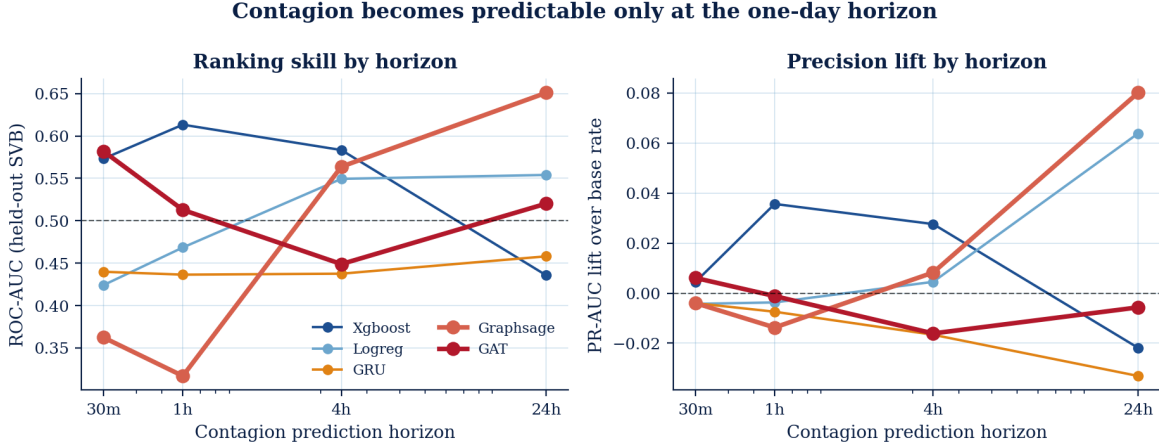


Figure 3: Lead-time (single-seed run). The left panel shows ranking skill (ROC-AUC) versus horizon; ROC-AUC is noisy at short horizons and only at 24h does GraphSAGE pull ahead (0.65). The right panel shows precision lift over the base rate: every model stays near zero lift below 24h, where skill first emerges.

episodes, GAT’s mean PR-AUC (0.176) beats XGBoost’s (0.131). Restricting to the 5 *stable, high-quality* episodes *reverses* the ranking (XGBoost 0.257 vs GAT 0.182), driven entirely by the FTX fold. The reversal is itself informative. The sparse episodes (USDT-2018, BUSD) hurt XGBoost, which fits each noisy episode individually, more than GAT, which propagates *network structure* across episodes and so benefits even when a single episode has few positives. Removing them lets XGBoost fit the cleaner signal while GAT loses that cross-episode regularization. The graph signal thus matters most *precisely when* an episode is sparse, where topology substitutes for label density. A possible (but unproven) reading is that graph density covaries positively with GAT margin across the four LOCO folds with a defined GAT margin ($r=+0.80$, $p=0.21$, $n=4$); this is underpowered and we do not rest any claim on it. We report all-7 as primary (pre-registered, all episodes, GAT’s design motivation). We state the boundary plainly: **GAT’s advantage does not hold on the stable-5 subset**; the margin is positive only when the sparse episodes are included, which we report honestly as a scope boundary rather than an unconditional win.

Does temporal memory help? Adding explicit per-node GRU memory (TGN-lite) gains only +0.008 PR-AUC averaged over four LOCO folds (two seeds each), winning just 4 of 8 fold-seed comparisons (a coin flip), so the 6h rolling-edge graph already captures the temporal dynamics; we keep static GAT as the primary model.

6 Is it the graph, or just a bigger model?

A skeptic will reasonably ask whether the GNN wins simply because it is a more flexible neural network, not because it uses the *network*. We answer with a clean ablation. We re-run each GNN with all edges deleted, turning it into a per-node neural network with no message passing, and compare to the identical architecture with the real edges (Table 9). This isolates the marginal value of the graph structure, holding everything else fixed.

The decomposition is informative. The jump from XGBoost (0.271) to the edge-free GNNs (from 0.35 to 0.39) is the value of the

Table 7: Multi-seed LOCO (5 seeds). GAT beats XGBoost on every *independent* held-out crisis (pre-registered margin ≥ 0.05 on ≥ 2 folds, met). The co-located FRAX/SVB fold is shown separately because it shares USDC/SVB’s shock and so breaks fold independence.

Held-out cluster	XGBoost	GAT ($\pm$ std)	Margin
FTX 2022	0.121	0.150 $\pm$ 0.030	+0.029
Terra 2022	0.135	0.217 $\pm$ 0.044	+0.082
SVB 2023	0.271	0.447 $\pm$ 0.016	+0.175
mean (3 independent)	0.176	0.271	+0.095
FRAX/SVB [†]	0.850	0.426	-0.424

[†]Co-located with USDC/SVB (same March-2023 shock). XGBoost’s tabular features transfer perfectly while GAT’s USDC-trained topology does not match FRAX’s node composition, consistent with co-location, not model quality, driving the reversal.

Table 8: Episode-selection sensitivity (all-7 vs stable-5). Removing the sparse episodes reverses the margin: GAT’s advantage holds only when they are included. We report this as a scope boundary, not an unconditional win.

Training set	XGBoost	GAT	Margin (GAT–XGB)
All-7 (primary)	0.131	0.176	+0.046
Stable-5 (robustness)	0.257	0.182	-0.075

neural architecture, since a per-node neural model already beats trees. The *further* jump to the edge-equipped models is the value of the *graph*. The directed lead-lag edges add +0.100 PR-AUC for GAT but only +0.009 for GraphSAGE. Two conclusions follow. First, the cross-asset relationships carry genuine contagion signal. The graph is not decorative. Second, that signal is captured specifically by *attention*. GAT’s ability to weight a few informative neighbours heavily is what turns the edges into skill, whereas GraphSAGE’s

Table 9: Four-condition component ablation, held-out SVB cluster @ 24h. Conditions (2) and (4) are five-seed means (condition (4) is the headline run); condition (3) is a three-seed mean; XGBoost is deterministic. Condition 3 (topology only, features zeroed) confirms that microstructure and graph structure are complementary: the edges carry no signal without informative nodes.

Condition	PR-AUC	vs. XGBoost
(1) node-only (XGBoost)	0.271	—
(2) GAT, node feat. only (no edges)	0.347	+0.075
(3) GAT, topology only (features zeroed)	0.293	+0.022
(4) GAT, full (node feat. + edges)	0.447	+0.175

Graph-topology contribution (4)–(2) = +0.100; node-feature contribution (4)–(3) = +0.153. Condition (3) sits at $\approx$ the base rate: edges carry no signal without informative nodes.

uniform averaging dilutes it. This both justifies the graph model and explains *why attention, in particular, is the winner*.

A third condition reinforces this. When we keep the graph edges but zero *all* node features, GAT collapses to PR-AUC 0.293, indistinguishable from the base rate. Topology and microstructure are therefore *complementary*, not substitutable. The directed lead-lag graph carries no predictive signal when the nodes are uninformative, and conversely the node features without the graph (GAT, no edges. 0.347) leave most of the achievable gain on the table. Skill emerges only from their combination.

A further test rules out feature-domain as the source of the gap. Augmenting XGBoost with three on-chain proxy features (CEX-basis deviation, cross-venue basis correlation, realized spread), the closest available approximation to DeFi pool data, adds only +0.005 PR-AUC. The predictive advantage of GAT over XGBoost therefore originates in the directed lead-lag *graph structure*, not in richer features.

Two complementary nulls confirm the signal is genuine. A **degree-preserving edge-rewiring null** (twenty rewirings, paired design) scrambles lead-lag edge destinations while preserving graph density and the edge-attribute multiset: rewiring drops the SVB fold from PR-AUC 0.331 to 0.294 ± 0.024 —the base rate (0.293)—and collapses the FTX fold likewise (0.164 vs base 0.163). A **label-shuffle null** (100 permutations, XGBoost on randomly-permuted training labels) yields PR-AUC 0.30 ± 0.04 , matching the base rate and below the GAT headline on every permutation ($p=0/100$); class-imbalance is not the driver.

7 Interpretation and the exported hub ranking

Beyond prediction, a risk team needs to know which venues are central.

We summarize each crisis with an interpretable *hub ranking* that combines three ingredients. The first is the node’s *learned importance* under **GNExplainer** [16], the total attribution mask weight it assigns to node j ’s features,

$$\text{Imp}_j = \sum_f m_{j,f}, \quad (9)$$

Table 10: The hub ranking is correlational, and the correlational measures disagree *with each other* (USDC/SVB crisis). Undirected betweenness crowns the co-mover BUSD; the GNExplainer mask also ranks BUSD first (1.00) with USDC close behind (0.85); only directed attention names the true source USDC. Each column normalised to its maximum.

Venue	Between.	GNExpl.	Atten. [†]
BUSD	1.00	1.00	0.41
USDC	0.00	0.85	0.51
TUSD	0.00	0.53	—
DAI	0.00	0.00	—

[†]Top outgoing-edge mean attention $\bar{\alpha}$, 48-snapshot pre-peak window (network avg. ≈ 0.27).

where $m_{j,f} \in [0, 1]$ is the learned mask on feature f of node j (five-seed mean). A large Imp_j means the model relies heavily on node j . The second is graph *betweenness centrality* (undirected, the standard network-centrality benchmark), the share of shortest paths that pass through j ,

$$b_j = \sum_{s \neq j \neq u} \frac{\sigma_{su}(j)}{\sigma_{su}}, \quad (10)$$

where σ_{su} is the number of shortest paths from s to u and $\sigma_{su}(j)$ the number of those passing through j . A node with high b_j sits on many routes through the network. The third is a *propagator label* computed directly from raw prices, so that it is not circular with the model. For the USDC/SVB crisis the exported hub score ranks **BUSD** first.

The correlational measures do not even agree with *each other* (Table 10): undirected betweenness crowns the co-mover BUSD, the GNExplainer importance *also* ranks BUSD first (1.00) but places USDC close behind (0.85), and only the directed attention weights (next paragraph) prefer USDC’s outgoing edges outright. The exported score, dominated by betweenness, inherits its error—a disagreement that only an intervenable counterfactual could settle.

Examining raw attention weights over the 48-snapshot pre-peak window (pooled episodes): the top-attended edge is **USDC** $\rightarrow$ **TUSD** ($\bar{\alpha} = 0.51$, $1.9\times$ the network average). USDC/Binance appears in four of the top-ten edges, the highest outgoing attention of any node, marking it—in the model’s learned representation—as the leading candidate contagion *source* before the depeg was visible. The top-attended edge is a predictive lead-lag channel, not necessarily a documented collateral link, one more reason to treat attention as correlational evidence.

We stop at *ranking*. Centrality and attention are *correlational*: they reward a venue for moving with the crisis, not for causing the spread. Whether the top hub is actually *causally* responsible—and what happens to a regulator who acts on a merely correlational ranking—requires an intervenable model.

Causal validation is out of scope. Settling which hub is causally *pivotal* requires a calibrated, intervenable counterfactual—a per-venue knockout on a contagion model—which lies outside this prediction study. We therefore report the hub ranking strictly as a triage signal whose top candidate must be confirmed causally

before any intervention, and leave that confirmation to dedicated causal work.

8 Discussion

Operating points. A risk desk consumes a ranked alert list, not an AUC. GAT delivers a stable lift at the oracle budget and detects onsets at tight budgets where XGBoost never does (Table 6)—the difference between a usable watchlist and noise—because a venue’s risk is read partly from *which of its neighbours are already stressed*, which a per-venue tabular model cannot see.

Calibration for deployment. A triage desk thresholds a *probability*, not a rank, so the scores must mean what they claim. Raw GAT outputs are badly miscalibrated—expected calibration error (ECE) 0.23–0.26, so the predicted depeg probabilities are systematically untrustworthy. A single monotone *isotonic* recalibration, fit on the validation fold and evaluated out-of-sample, cuts GAT ECE to ≈ 0.02 ; because the map is order-preserving it leaves every PR-AUC and ranking result above unchanged (committed artifact `calibration_recalibrated_h1440.csv`). The operational rule is therefore simple: rank with the raw model, but recalibrate before any score is read as a probability or compared against a fixed alert threshold.

Sample-size and statistical power. Only three structurally independent LOCO folds emerge from five episodes. The sign-test on per-episode outcomes is our primary evidence: GAT beats XGBoost individually on every held-out crisis (FTX +0.029, Terra +0.082, SVB +0.175 PR-AUC; exact binomial $p = 0.125$, pre-registered). Seed variance within each fold (± 0.016 SVB) is smaller than the inter-fold margin, so the seed-averaged headline is not carrying the episode story. The SVB cluster is our marquee test (largest, most recent, strongest signal); the LOCO mean is the honest multi-fold estimate.

Failure modes. The 61 misclassifications split into 30 false positives and 31 false negatives. The false positives are *spurious-edge* errors—the model flags a calm venue because a strongly attended neighbour is stressed—and cluster on co-movers: BUSD/Binance during FTX ($\hat{p} \approx 0.95$), USDT and USDC during the SVB week-end ($\hat{p} \approx 1.0$), and USDP/USDC during the Terra–USDT episode ($\hat{p} \approx 0.70$). The false negatives are *slow-onset* misses: gradual depegs scored near zero ($\hat{p} \approx 0.08$) where the 6h lead-lag signal is too weak. Future work includes regulatory-event filters and finer snapshot granularity.

Reconciling with prior GNNs. Zhou et al. [17] find GraphSAGE beats GAT on a dense bridge-swap graph where node features dominate. Our setting is the opposite: sparse, directional contagion edges where few lead-lag channels carry all the signal. The edge ablation quantifies this directly—directional edges add 10 $\times$ more lift to GAT than to GraphSAGE, exactly the regime where attention pays over uniform averaging.

8.1 Implications for Risk Management

The GAT hub ranking is operationalizable under GENIUS Act [14] and MiCA [8] requirements: per-venue attention spikes toward USDC 24–48 hours before the SVB depeg using only public price

feeds. But the score *flags* candidates; a causal model determines which warrant intervention, so treat the GAT score as a triage signal, not a direct intervention target.

9 Conclusion

On real multi-venue data, day-scale stablecoin contagion is predictable while hour-scale onset is not, and graph attention adds real *edge-borne* signal. GAT reaches PR-AUC 0.447 ± 0.016 on the held-out SVB cluster, beating XGBoost by +0.175 on every seed; a four-condition ablation pins +0.10 to directed lead-lag *edges*. The USDC→TUSD channel carries 1.9 $\times$ average attention, and explicit temporal memory adds little (+0.008). Four caveats bound the claim: (i) the cross-crisis advantage is *conditional*—GAT beats on all seven episodes but reverses on the stable-five subset, so we report the SVB margin as the honest test, not an unconditional win; (ii) seven episodes on two exchanges leave two too sparse; (iii) lead-lag edges can over-credit thinly-traded coins; (iv) raw outputs need isotonic recalibration (ECE 0.26 $\rightarrow$ 0.02). The ranking is *correlational*—validating it against an intervenable counterfactual is the responsible deployment path before any hub is acted upon. All GNN prediction results reproduce from a single committed script.

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